9th Feb 2026 | DA6360

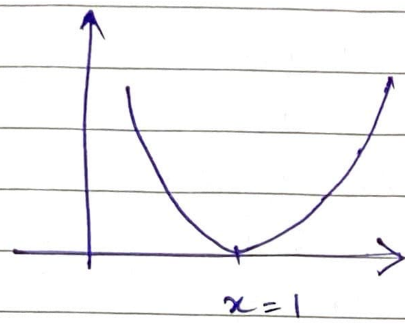
Worksheet soon

Sat before 16th March

3 hours - Morning

→ Nec Eq. of FONC

- $f(x) = (x-1)^2$



$f'(x) = 2(x-1)$

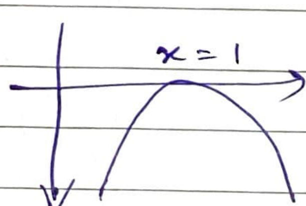
x^* is local minima

$\Rightarrow f'(x_*) = 0$

@ $x = 1$

$f'(1) = 0$

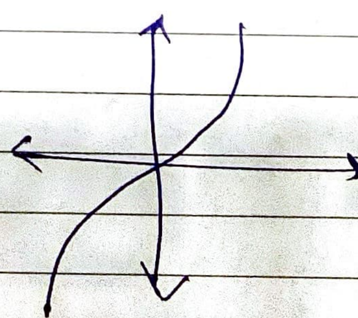
- $f(x) = -(x-1)^2$



$f'(x_*) = 0$

$\Rightarrow x_*$ is a local minima

- $f(x) = x^3$



$f'(x) = 3x^2$

$f'(0) = 0$

neither min nor max.

any x_* where $f'(x_*) = 0$ is known as stationary pt.

→ Second Order Necessary Condition

assume → f, f', f'' (continuous)

If x^* is a local minima, then $f''(x_*) \geq 0$.

Proof – Suppose not, i.e. say $f''(x_*) < 0$

$\Rightarrow$ we can prod. a better pt.

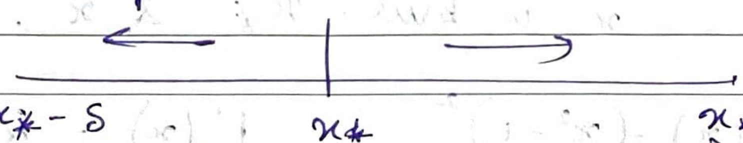
(1) $f''(x_*) < 0$

(2) $\exists$ a $\delta > 0$ s.t

$x \in (x_* - \delta, x_* + \delta)$ &

$f''(x_* < 0)$

(3) Pick any $x \in (x_* - \delta, x_* + \delta)$



$$f(x) = f(x_*) + \underbrace{f'(x_*)}_{= 0 \text{ FONC}} (x - x_*) + \frac{1}{2} \underbrace{f''(\bar{x})}_{<0} \underbrace{(x - x_*)^2}_{+ve}$$

$\bar{x}$ lies btw. x_* & x : $\downarrow$ does not matter

$\rightarrow$ <u>Second Order Sufficient Condition</u> (SOSC)

If (i) $f'(x_*) = 0$

(ii) $f''(x_*) > 0$ $\Downarrow$

then x_* is a strict local minimiser.

$$f(x) = f(x_*) + f'(x_*)(x - x_*) + \underbrace{f''(\bar{x})}_{>0} \times \underbrace{\frac{(x - x_*)^2}{2}}_{>0}$$

$x \in (x_* - \delta, x_* + \delta)$

$f(x) > f(x_*)$

not x_* is min.
$\Rightarrow f \geq 0$

- x_* is a strict local minimiser (sufficient & necessary)

$\Updownarrow$ iff.

first non-0 term in $\{f^{n'}(x_*)\}$ is >0.
& occurs at min n.

$$f(x) = f(x_*) + \underbrace{f'(x_*)}_{=0}(x - x_*) + \underbrace{\frac{f''(x_*)}{2}}_{=0}(x - x_*)^2$$
$$+ \dots + \underbrace{\frac{f^{(2k-1)'}(x_*)}{2k+1}}_{\neq 0}(x - x_*)^{2k+1} + \underbrace{\dots}_{>0}$$

$\bar{x}$ is btw. x_* & x.

Eg. $f(x) = (x^2-1)^2$, $f'(x) = 2(x^2-1)\,2x$
at $x = 0, -1, +1$ $f'(x) = 0$

$f''(x) = 12x^2 - 4$

$f''(0) = -4 < 0$ local maxima
$f''(\pm 1) = 12 - 4 = 8 > 0$ local minima

Eg. $f(x) = (x^2-1)^3$, $f'(x) = 3(x^2-1)^2\,2x$
$= 6x(x^4+1-2x^2)$
stationary pts. $x = 0, +1, -1$ $6x^5 - 12x^3 + 6x$

$f''(x) = 6(x^2-1)^2 + 12x(x^2-1)\,2x$

$= 30x^4 - 36x^2 + 6$

$f''(0) = 6 > 0$ local min is
$f''(-1) = 30 - 36 + 6 = 0$ probe further
$f''(+1) = 0$ "

$f'''(x) = 120x^3 - 72x$

$f'''(+1) = 120 - 72 = 48 > 0$

$f'''(-1) = -120 + 72 = -48 < 0$

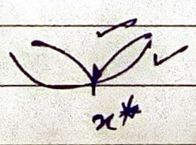
(in higher dim., you will get saddle pt.)

it 1 direc^n increasing, it other direc^n it decreases

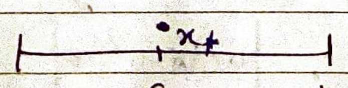
⇒ **Optimizing in 1-D**

- $f(x)$ is a unimodal func^n in the interval $[a, b]$ if it is
  - → strictly decreasing to the left of x_* ($a \leq x_1 < x_2 \leq b$, $x_2 < x_* \Rightarrow f(x_1) > f(x_2)$)
  - → strictly increasing to the right of x_*

    $x_1 > x_* \Rightarrow f(x_2) > f(x_1)$



- Bisec^n method → we have info. of $f'(x)$



$a' = a$, $b' = b$

for $k = 1, 2 \ldots$

$c^k = \dfrac{a^k + b^k}{2}$ (center)

if $f'(c_k) > 0$

$b^{k+1} \leftarrow c_k$

if $f'(c_k) < 0$

$a^{k+1} \leftarrow c_k$

$f'(c_k) = 0 \Rightarrow x_* = c_k$

not helpful in high dimensions.

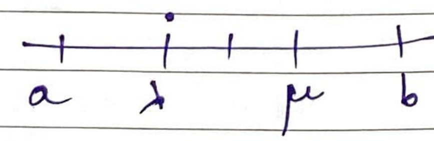
$$len([a^{k+1}, b^{k+1}]) = \frac{len([a,b])}{2^k}$$

- 0th Oracle - Dichotomous Search

$a^1 = a, \quad b^1 = b$ for $k = 1, 2, \ldots$ $\varepsilon^k = \varepsilon(len[a^k, b^k])$

$$\mu^k = \frac{a^k + b^k}{2} + \varepsilon^k$$

$$\lambda^k = \frac{a^k + b^k}{2} - \varepsilon^k$$



it should dec. then increase.

if $f(\mu^k) > f(\lambda^k)$

$b^{k+1} = \mu^k$

if $f(\mu^k) < f(\lambda^k)$

$a^{k+1} = \lambda^k$

if $f(\mu^k) = f(\lambda^k)$

$b^{k+1} = \mu^k$
$a^{k+1} = \lambda^k$

$k=1$, $len([a^2, b^2]) \leq \frac{len([a,b])}{2} + \varepsilon$

normalise $[a,b] = 1$, factor of $0.5 + \varepsilon$

$(0.5+\varepsilon)^k \rightarrow$ shrink